Hanjun Park

Department of Industrial, Manufacturing, & Systems Engineering
Texas Tech University, Box 43061, Lubbock, TX 79409-3061
hanjpark@ttu.edu   Lab webpage: https://hanj321.github.io

ACADEMIC APPOINTMENT

Assistant Professor *09/2025–Present*
Texas Tech University
- Department of Industrial, Manufacturing, and Systems Engineering

Graduate Research Assistant *2019–2025*
Occupational Ergonomics and Biomechanics Laboratories
- Faculty Advisors: Maury A. Nussbaum & Divya Srinivasan

Graduate Research Assistant *2012–2014*
Life Enhancing Technology Laboratory
- Faculty Advisor: Woojin Park

EDUCATION

Ph.D., Industrial and Systems Engineering *08/2025*
Virginia Tech, Blacksburg, VA
- Graduate Certificate in Data Analytics

M.S., Industrial Engineering *08/2014*
Seoul National University, Seoul, Korea

B.S., Industrial and Systems Engineering *12/2011*
Georgia Institute of Technology, Atlanta, GA

PUBLICATIONS

Peer-Reviewed Journal Publications (* corresponding-author)

[*J*.8] Santos, F., Son, C.*, **Park, H.,** Xue. C, Yang, J., & Liang, N. Design, fabrication, and testing of an ergonomic intervention for a virtual reality headset using 3D printing. *Ergonomics*. *Accepted*

[*J*.7] **Park, H.,** Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2026). Adaptation to a whole-body powered exoskeleton: Human-exoskeleton coordination during load-handling tasks. *Annals of Biomedical Engineering*. 1-15. doi:10.1007/s10439-026-04025-9

[*J*.6] **Park, H.*,** & Nussbaum, M.A. (2025). Muscle synergy analysis during pseudo-static and dynamic overhead tasks with arm-support exoskeletons. *Journal of Biomechanics*, 113135. doi:10.1016/j.jbiomech.2025.113135

[*J*.5] Ransing, V., Park, J., Ye, Y., **Park, H.,** Kim, S., Du, J., & Srinivasan, D.* (2025). Efficacy of virtual reality-based approach for users to understand the potential benefits and limitations of using exoskeletons. *Human Factors*, 67(11), 1136-1151. doi:10.1177/00187208251346627

[*J*.4] **Park, H.,** Noll, A., Kim, S., & Nussbaum, M.A.* (2025). Passive arm-support and back-support exoskeletons have distinct phase-dependent effects on physical demands during cart pushing and pulling: An exploratory study. *Applied Ergonomics*, 126, 104510. doi:10.1016/j.apergo.2025.104510

[*J*.3] **Park, H.,** Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2023). A pilot study investigating motor adaptations when learning to walk with a whole-body powered exoskeleton. *Journal of Electromyography and Kinesiology*, 69, 102755. doi:10.1016/j.jelekin.2023.102755

[*J*.2] **Park, H.,** Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2022). Effects of using a whole-body powered exoskeleton during simulated occupational load-handling tasks: A pilot study. *Applied Ergonomics*, 98, 103589. doi:10.1016/j.apergo.2021.103589

[*J*.1] **Park, H.,** Park, W.*, & Kim, Y. (2014). Manikin families representing obese airline passengers in the U.S. *Journal of Healthcare Engineering*, 5(4), 479-504. doi:10.1260/2040-2295.5.4.479

Journal Publications in Progress

[*U*.1] **Park, H.*,** & Nussbaum, M.A. Characterizing adaptation to using exoskeletons: A narrative review and assessment framework for occupational contexts. *Applied Ergonomics*. *Under Review*

[*U*.2] Santos, F., Son, C.*, **Park, H.,** Xue. C. Yang, J., & Liang, N. Biomechanical effects of virtual reality headset support and task demands on the neck. *IISE Transactions on Occupational Ergonomics and Human Factors. In Revision*

[*U*.3] **Park, H.*,** & Nussbaum, M.A. Perceptual adaptations to back-support exoskeletons: evidence from measures of workload and discomfort. *In Preparation*

[*U*.4] Lee, W., Kwon, S., **Park, H.***, Hwang, D.* Data-driven identification of troubleshooting patterns in 3D printing: a text mining approach. *In Preparation*

[*U*.5] Choi, W.Y., **Park, H.**, Jeong, Y.*, Hwang, D.* How motion parameters influence perceived usability in autonomous personal wheelchair: A comparative evaluation with younger and older adults. *In Preparation*

Peer-Reviewed Conference Proceedings

[*C*.6] **Park, H.,** Kim, S., Nussbaum, M.A., & Srinivasan, D. (2022). Changes in kinematics and muscle activity when learning to use a whole-body powered exoskeleton for stationary load handling. *Proceedings of the 66th Human Factors and Ergonomics Society Annual Meeting, 66(1), 273–274.*

[*C*.5] **Park, H.,** Lee, Y., Kim, S., Nussbaum, M.A., & Srinivasan, D. (2021). Gait kinematics when learning to use a whole-body powered exoskeleton. *Proceedings of the 65th Human Factors and Ergonomics Society Annual Meeting, 65(1), 1367–1368.*

[*C*.4] **Park, H.,** Kim, S., Lawton, W., Nussbaum, M.A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual handling. *Proceedings of the 64th Human Factors and Ergonomics Society Annual Meeting, 64(1), 888–889.* **- Won Best Student Paper Award at the Occupational Ergonomics Technical Group**

[*C*.3] Jung, J., **Park, H.,** Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2014). A review on interaction techniques in virtual environments. *Proceedings of the 2014 International Conference on Industrial Engineering and Operations Management (pp. 1582-1590).*

[*C*.2] Beck, D., Hwang, D., Son, M., Jung, J., **Park, H.,** Park, J., & Park, W. (2013). A Review on Techniques for Evaluating Physical Stresses Associated with the Use of Various Input Devices. *Proceedings of the 2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea.*

[*C*.1] Jung, J., **Park, H.,** Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2013). A Review on Object Manipulation Techniques in Virtual Environments and their Performance Metrics. *Proceedings of 2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea.*

HONORS & AWARDS

- Best Experimental Paper Award, OETG at HFES *2024*
- Best Student Poster Presentation Award, ASSP Regional Conference *2023*
- IISE Future Faculty Fellow (3F) Program Awardee *2023–2024*
- NIOSH Training Fellow, Virginia Tech *2023–2025*
  - Fellowship includes monthly stipend, tuition, and required fees
- ISE Graduate Travel Award ($900), Virginia Tech *2022*
- Best Student Paper Award, OETG at HFES *2020*
- VT HFES Student Chapter Research Competition Award *2020*
- Grado Department of ISE Fellowship, Virginia Tech *2018*
  - Fellowship includes monthly stipend, tuition, and required fees
- Brain Korea 21 Scholarship, Seoul National University *2014*
- Lecture & Research Scholarship, Seoul National University *2013*

PRESENTATIONS

Peer-reviewed Conference Oral Presentations

[*O*.10] Santos, F., & Son, C., **Park, H.,** (2026). A 3D-printed shoulder support to reduce neck strain during virtual reality device use. *30th Applied Ergonomics Conference, Arlington, TX.*

[*O*.9] Santos, F., **Park, H.,** & Son, C. (2025). A 3D-printed shoulder support to reduce neck strain during prolonged virtual reality (VR) device use: pilot findings. *69th Human Factors and Ergonomics Society Annual Meeting, Chicago, IL.*

[*O*.8] **Park, H.,** Ojelade A., Kim, S., & Nussbaum, M.A. (2024). Effects of using arm-support exoskeletons on muscle coordination during a pseudo-static overhead task: preliminary analysis of muscle synergy. *68th Human Factors and Ergonomics Society Annual Meeting, Pheonix, AZ.* **- Won Best Experimental Paper Award at the Occupational Ergonomics Technical Group.**

[*O*.7] **Park, H.**, Noll, A., Kim, S., & Nussbaum, M.A. (2023). Effects of using passive back- and arm-support exoskeletons for cart pushing and pulling. *67th Human Factors and Ergonomics Society Annual Meeting, Washington, DC*.

[*O*.6] Noll, A., Nussbaum, M.A., Kim, S., & **Park, H.** (2023). Reliability Analysis of Task Performance and Subjective Measures for Assessment of Occupational Exoskeletons. *67th Human Factors and Ergonomics Society Annual Meeting, Washington, DC*.

[*O*.5] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D. (2022). Changes in kinematics and muscle activity when learning to use a whole-body powered exoskeleton for stationary load handling. *66th Human Factors and Ergonomics Society Annual Meeting, Baltimore, MD*.

[*O*.4] **Park, H.**, Lee, Y., Kim, S., Nussbaum, M.A., & Srinivasan, D. (2021). Gait kinematics when learning to use a whole-body powered exoskeleton. *65th Human Factors and Ergonomics Society Annual Meeting, Baltimore, MD*.

[*O*.3] **Park, H.**, Kim, S., Lawton, W., Nussbaum, M.A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual handling. *64th Human Factors and Ergonomics Society Annual Meeting*.

[*O*.2] Beck, D., Hwang, D., Son, M., Jung, J., **Park, H.**, Park, J., & Park, W. (2013). A Review on Techniques for Evaluating Physical Stresses Associated with the Use of Various Input Devices. *2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea*.

[*O*.1] Jung, J., **Park, H.,** Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2013). A Review on Object Manipulation Techniques in Virtual Environments and their Performance Metrics. *2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea*.

Poster Presentations

[*P*.3] Santos, F., Son, C., **Park, H.,** Xue. C, Yang, J., & Liang, N. Design, fabrication, and testing of an ergonomic intervention for virtual reality headset using 3D printing. (2026). *WCOE Research Day, Texas Tech University, Lubbock, TX, February*.

[*P*.2] **Park, H.,** Ojelade, A., Kim, S., & Nussbaum, M. A. (2023). Effects of different arm-support exoskeletons on muscle activation patterns during a pseudo-static overhead task - preliminary analysis of muscle synergy. *37th American Society of Safety Professionals (ASSP) Region VI Professional Development Conference (PDC) in Myrtle Beach, South Carolina, September*. **- Won Student Presentation Award**

[*P*.1] **Park, H.,** Kim, S., Nussbaum, M. A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual material handling. *2020 Annual HFES/INFORMS Poster Competition, Virginia Tech, VA, October*. **- Won Student Poster Presentation Award**

Invited Seminars

[*S*.3] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Psychological Sciences, Texas Tech University, October*.

[*S*.2] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Industrial, Manufacturing, and Systems Engineering, Texas Tech University, April*.

[*S*.1] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Industrial and Systems Engineering, Kennesaw State University, March*.

Invited Panel Talk

[*T*.1] **Park, H.** (2025). From Student to Professor: Exploring Careers in Occupational Ergonomics. Panel Member. *Occupational Ergonomics Technical Group of HFES, September*.

TEACHING EXPERIENCE

Instructor of Record

Texas Tech, Dept. of Industrial, Manufacturing, & Systems Engineering

- IE 5304: Biomechanics & Work Physiology (n = 11) — *Spring, 2026*

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 4624: Work Physiology (n = 36) — *Fall, 2023*
- ISE 3614: Human Factors Engineering and Ergonomics (taught online, n = 7) — *Summer, 2023*

Guest Lecturer

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 3624: Industrial Ergonomics — *Spring, 2023*
  "NIOSH Lifting Equation and Controls" (n = 100)
- ISE 4644: Occupational Safety and Hazards Control — *Fall, 2018*
  "Review on Hazard Analysis and Prevention" (n = 20)

Graduate Teaching Assistant

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 3624: Industrial Ergonomics — *Spring, 2019*
- ISE 4624: Work Physiology — *Spring, 2019*
- ISE 4644: Occupational Health and Hazards Control — *Fall, 2018*

PROFESSIONAL EXPERIENCES

Software Engineer, *Military duty* — *10/2014–03/2018*
Kornic Glory, Seoul, Korea

Research Intern — *06/2011–08/2011*
LG Electronics, Seoul, Korea

ADVISING & MENTORING

Dissertation/Thesis Committee Member

- Daniel Leibman, TTU (PhD, Psychology) Chair: Heesun Choi (*2025–Present*)
- Armina Rahman, TTU (PhD, IMSE) Chair: Changwon Son (*2025–Present*)
- Felipe Santos, TTU (PhD, IMSE) Chair: Changwon Son (*2025–Present*) [*J*.8], [*O*.10], [*O*.9], [*U*.2]

Research Mentoring

Voluntary Postdoc Position

- Leonardo Wei, Texas Tech (PhD, ISE) (*01/2026–Present*)

M.S. Students

- Delaney Sheppard, Texas Tech (BIOE) (*02/2026–Present*)
- Alex Noll, Virginia Tech (ISE) (*08/2022–08/2024*) [*J*.4], [*C*.8], [*C*.7]

Undergraduate Students

- Enriquez Bazoalto Kaylee, Virginia Tech (ISE) (*05/2025-08/2025*)

PROFESSIONAL SERVICE ACTIVITIES

Service to Profession

Session Chair, the 69th HFES Annual Meeting, Atlanta, GA *2025*

Archival Journal

- Reviewer, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* *2026–Present*
- Reviewer, *Scientific Reports* *2026–Present*
- Reviewer, *PLOS One* *2026–Present*
- Reviewer, *IJIE: Theory, Applications, and Practice* *2026–Present*
- Reviewer, *International Journal of Mechanical System Dynamics* *2026–Present*
- Reviewer, *IEEE Transactions on Human-Machine Systems* *2025–Present*
- Reviewer, *Ergonomics in Design* *2024–Present*
- Reviewer, *IISE Transactions on Occupational Ergonomics and Human Factors* *2023–Present*
- Reviewer, *Human Factors and Ergonomics Society Proceedings* *2021–Present*

Service to Institution

Texas Tech Regional Science Fair (K-12) Special Award Judge *2026*

Texas Tech Regional Science Fair (K-12) Ribbon Judge *2026*

Dean's Representative, Dissertation Defense of Yves Valentin
(MS in Human Factors Psychology, TTU Department of Psychological Sciences) *2026*

TTU Graduate Academic Advisor *2025–Present*

VT Chapter of the Human Factors and Ergonomics Society (HFES)

- Officer: *Webmaster* *2020–2022*

National Biomechanics Day

- *Led lab tours and biomechanics activities for local high school students* *2020–2025*

VT Center for the Enhancement of Engineering Diversity (CEED) TechGirls Summer Camp

- *Led hands-on biomechanics activities* — *2019*

VT College of Engineering Alumni Outreach Event

- *Led lab tours for alumni* — *2019*

PROFESSIONAL AFFILIATIONS

Member, Institute of Industrial and Systems Engineers (IISE) — *2023–Present*

Member, Human Factors and Ergonomics Society (HFES) — *2020–Present*

Steering Committee member, K-Human Factors and Ergonomics Society (K-HFES) — *2021–Present*

REMARK

Dual Citizenship: U.S.A. and Republic of Korea